
Using MATSim for strategic planning: Insights from various projects at Swiss Federal Railways

Ihab Kaddoura, Thomas Hettinger, Merlin Unterfinger, Nicolas Dulex

Swiss Federal Railways (SBB)
Corporate Development, Strategy and Long-term Planning
Hilfikerstrasse 1, 3000 Bern, Switzerland

Extended Abstract

At Swiss Federal Railways (SBB), agent-based transport simulation has become an integral part of strategic and long-term planning. Building on the open-source *MATSim* (Multi-Agent Transport Simulation, www.matsim.org) framework, we apply a common modeling toolkit across a range of questions from the design of future nationwide service concepts, to the capacity of individual infrastructure elements, to the shift of freight from road to rail, and to the operational management of network disruptions. In this contribution we share insights from four recent projects that illustrate the breadth of this approach and the methodological lessons learned along the way.

Motivation and background. Switzerland's mobility landscape is set to change substantially towards 2050, when rising passenger and freight demand will require more efficient and flexible transport solutions. Meeting this growth is not a single challenge but a set of interrelated ones: the integral clock-face timetable is reaching its limits and may need to give way to more frequent, faster and more flexible (metro-like) service concepts; infrastructure layouts must be upgraded to provide reliable capacity where it is needed most; the available network capacity must be shared ever more carefully between passenger and cargo trains through smart slot planning; a growing share of inland freight should be shifted from road to rail; and the increasing volume of track renewal and maintenance work calls for smarter, passenger-aware construction-site planning. In this study, we present four projects, each of which addresses one of the challenges outlined above within the same simulation framework.

1. The future of railway: simulating future railway concepts. The first project deepens the exploration of long-term service concepts for the Swiss railway system. We quantified, using agent-based simulation, how Switzerland's mobility landscape might evolve towards 2050, when rising travel demand will require more efficient and flexible transport solutions. Specifically, we examined the effects of replacing the integral clock-face timetable with a metro-like system offering more frequent, faster and more flexible services, combined with regulatory measures. The findings indicated that the new railway concept and the regulatory measures each raise the modal share of public transport on their own, and that combining them yields the largest increase—while a detailed analysis showed that although most passengers benefit, some experience longer travel times. Building on these results, we now embed this exploration in an iterative design-and-evaluation framework: we simulate alternative supply concepts—including changes to the train schedule, stop patterns, first/last-mile connectivity and complementary regulatory measures—and assess their system-wide effects on modal shift, travel times, transfers and capacity load (vehicle and station occupancy). Designing in variants, rather than committing to a single blueprint, opens a more productive dialogue with decision makers and surfaces solutions that would otherwise remain hidden. The simulation outputs feed directly back into revising the rail supply, enabling continuous refinement and closer alignment with passengers' needs.

2. Capacity and robustness of infrastructure building blocks. The second project uses MATSim's *railsim* extension to investigate the operational performance and capacity of critical infrastructure elements in isolation. We evaluate a set of *building blocks* representing specific track-topology and operational variants: station layouts with one, two or three tracks per direction, and different crossing (junction) types—each combined with different operational schemes. Using Monte Carlo simulation with stochastic departure sampling, we derive a *timetable-flexible* measure of capacity that captures the robustness of a layout across a wide range of potential service patterns, rather than validating it against a single fixed schedule. The results reveal that reliable capacity is strongly influenced by the heterogeneity of the product mix and stopping frequency: adding a second platform track provides significant capacity gains, whereas a third track yields only marginal returns as the bottleneck shifts to adjacent single-track sections; similarly, grade separation resolves crossing conflicts but relocates the constraint to the downstream merging point. This provides an early yardstick for identifying where layout upgrades are required, before full network effects are accounted for.

3. Shifting inland freight from road to rail. The third project applies MATSim in the freight context to investigate the potential shift of Switzerland's inland freight transport from road to combined road/rail transport. Future freight demand for inland road transport is first calculated based on official governmental forecasting tools provided by ARE (Bundesamt für Raumentwicklung). MATSim is then used to model and evaluate different supply concepts: each TEU (twenty-foot equivalent unit) container is modeled as an agent that minimizes a predefined cost function by adjusting its mode of transportation, departure time and (intermodal) transport route. The simulated supply consists of the road network, the rail network, the cargo rail schedule, and the terminals where containers are transshipped between Heavy Goods Vehicles (HGV) and cargo trains. Going beyond existing approaches, dynamic queuing effects are explicitly represented for both systems: road segments and terminal cranes are modeled as capacity-constrained first-in-first-out queues, and trains have a limited capacity. The illustrative case study for 2050 reveals that combined road/rail transport offers great potential to reduce road traffic: switching from road to combined transport yields an average user cost reduction of 46%, and even without any improvement of the transit schedule or terminal capacities a significant trip share of 23% shifts to combined transport. Train and terminal capacities and the number of train departures per origin–destination relation emerge as decisive limiting factors, and a sensitivity analysis highlights how autonomous-driving cost effects could reshape the competitiveness of combined transport. This demonstrates how the same agent-based toolkit extends naturally from passenger to freight planning and to the joint use of network capacity by passenger and cargo trains.

4. MATSim for construction-site planning. The fourth project applies MATSim to the operational planning of network disruptions. For every pair of adjacent railway stations in Switzerland, we simulate an interruption of the connecting line and introduce a replacement bus service between the affected stations. Running the full agent-based demand simulation, we quantify the impact on railway users, including changes in demand flows, travel times and the number of transfers, as well as resulting overloads on trains and stations. This systematic, network-wide screening highlights which interruptions are particularly critical for passengers and where replacement concepts need to be strengthened, giving planners a data-driven basis for scheduling and mitigating construction work.

Methodological takeaways. Across these projects, several cross-cutting lessons emerged: plan under uncertainty by exploring multiple scenarios and presenting results incrementally; design in variants to uncover better solutions; leverage high-resolution simulation to enforce a holistic view and expose data inconsistencies; frame numbers carefully for top-management audiences by stating the reference metric and timeframe; and separate model development from model application to obtain an implicit "four-eyes" review that strengthens robustness.

The strategic planning efforts at SBB are ongoing, and the MATSim framework will continue to support upcoming project phases and conceptual ideas. The presented projects show that MATSim is now used across very different areas: strategic service design, infrastructure capacity, freight mode shift and operational disruption management, all within a single, consistent framework. Notably, AI-assisted development now lowers the barrier considerably, empowering colleagues at SBB who are not MATSim experts or at the beginner level to confidently work with and extend the simulation setup. Together, these factors demonstrate how a shared simulation framework delivers substantial value across a wide range of strategic, infrastructural and operational planning questions alike.